

# MarketMind AI — Project Notes

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## Autonomous Business Research Agent Workspace

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[ Autonomous Agentic Loop ] → [ Python Task Orchestration ]  
                                |  
[ Operational Cost Tracker ] → [ Human Approval Gate ] → [ Streamlit Workspace  
Dashboard ]
```

## Main Goal

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MarketMind AI is a workspace for an autonomous business research agent. Its job is to explore market signals and performance metrics on its own, turning a research question into a working set of findings without needing to be walked through every step by hand.

## How It Works

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The agent runs on native Python orchestration rather than a third-party agent framework, which lets it execute agentic loops — cycles of planning, acting, and reassessing — independently. Sitting behind these loops is a telemetry system that continuously tracks operational token costs, so the person overseeing the agent always knows what a given research run is costing as it happens, not just after the fact.

## Safety Control

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Because an autonomous agent can otherwise run ahead and take costly or consequential actions on its own, MarketMind AI includes an integrated human approval gate. Before any high-stakes task is executed — a paid data pull, a large-scale scrape, anything with real cost or consequence attached — the agent pauses and requires manual sign-off before proceeding. This keeps a person in the loop at exactly the moments that matter most, without requiring them to babysit the entire process.

## The Interface

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The workspace is built with Streamlit, giving the researcher a clear dashboard where market intelligence analytics — competitor comparisons, trend summaries, performance metrics — are laid out as they're produced, alongside the live cost tracker and any pending approval requests.